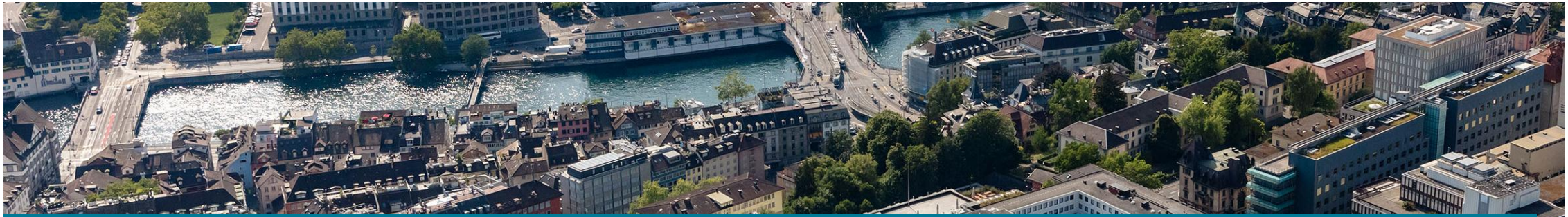
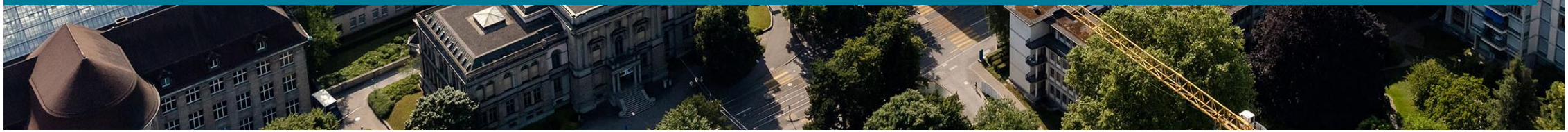
ETH zürich

RRE
Reliability & Risk Engineering

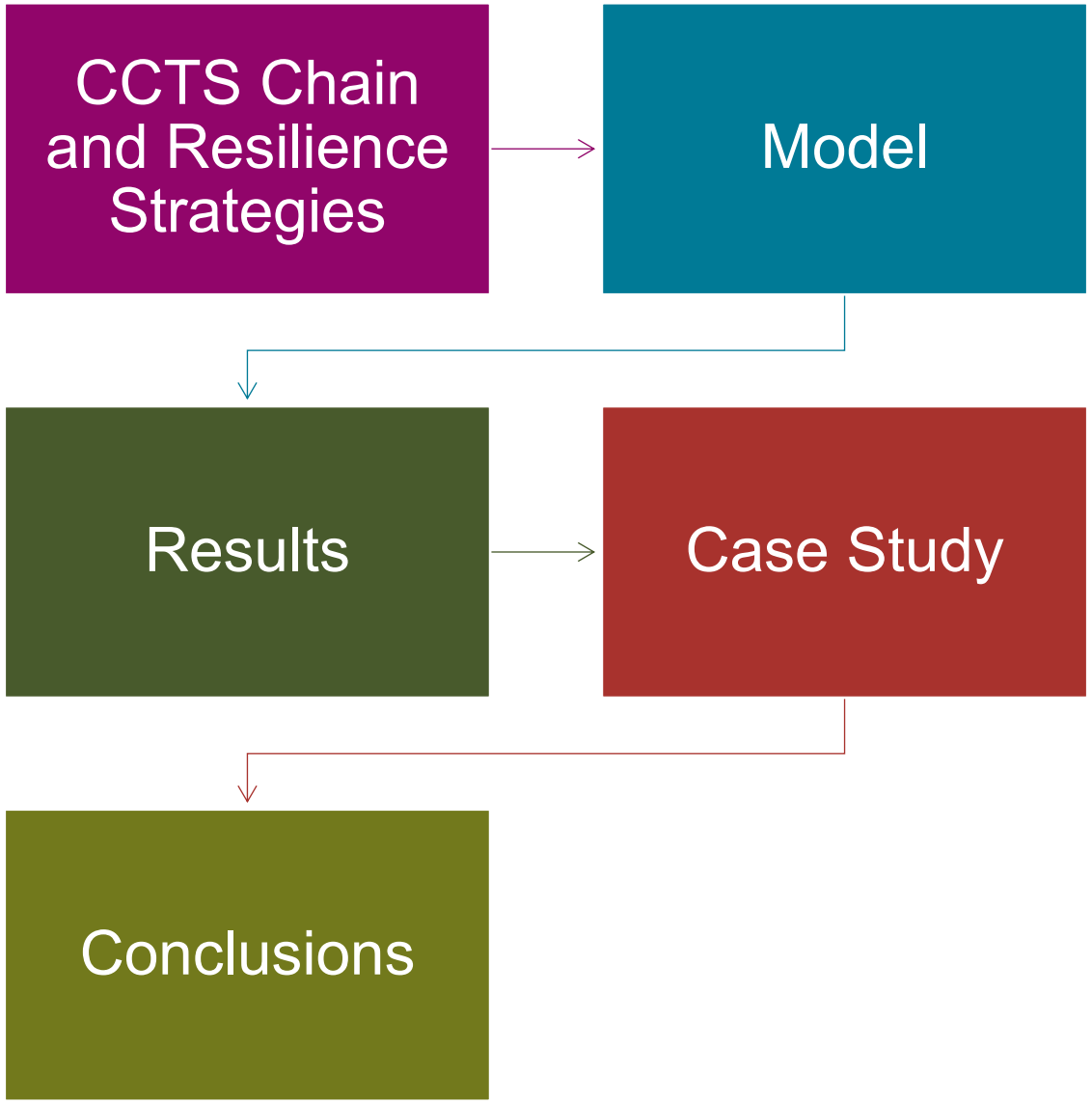


Techno Economic Assessment of CO_2 Transport Resilience in the CCTS Chain under Main Connection Failure Risk

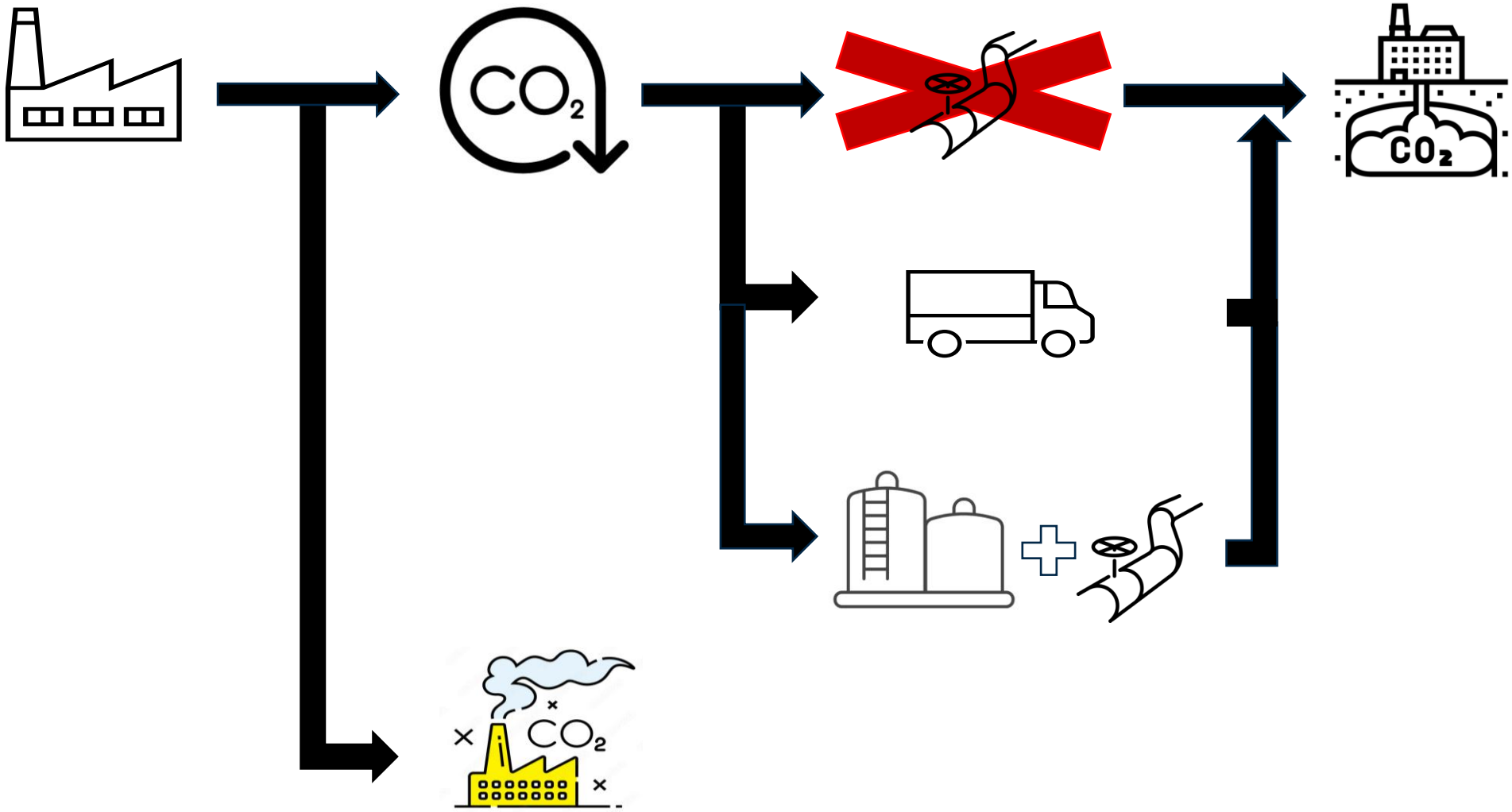
Constantinos Paschalis
Supervisors: Prof. Dr. Giovanni Sansavini
Johannes Burger
12 January 2026, Zurich



Agenda


CCTS Chain and Resilience Strategies
Model
Results
Case Study
Conclusions

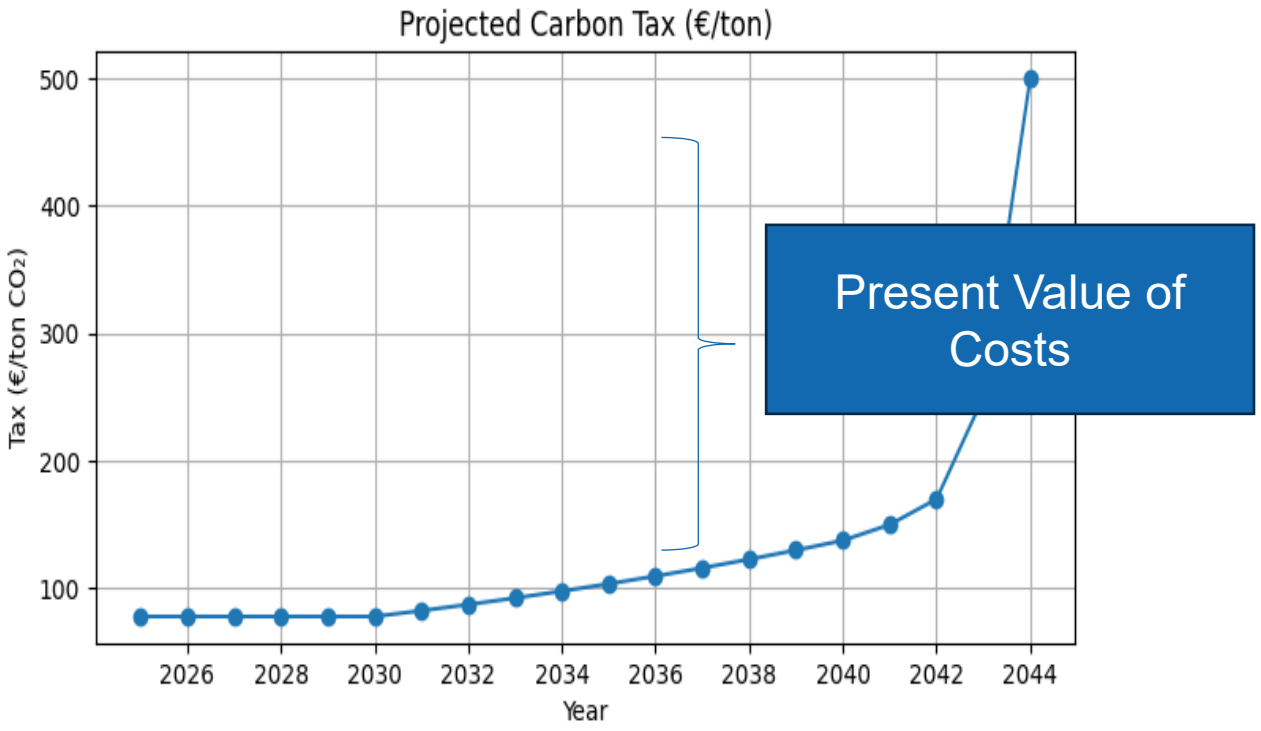
CCTS Chain and Resilience Strategies



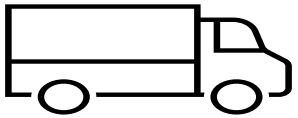
Model

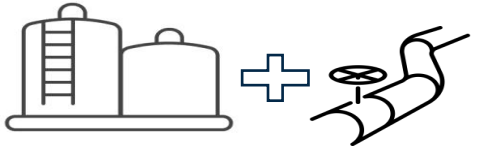
Input Parameters

- WtE power plant annual CO_2 emissions $[M_{CO_2}]$
- Length of main Connection [L]
- Buffer days for the main connection $[D_b]$
- Expected failure days in a year $[D_f]$
- Carbon price [T]



Model





Barges, ships and pipelines

Unitary cost (UC) expressions was taken from literature and operational fractions used for separating capital and operational costs

Trucks and trains:

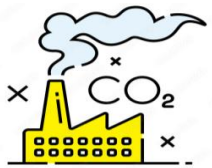
Mass dependent model created to remove the negative effect of low utilization
- Number of Trucks calculated and associated costs

Cost of temporary storage:

$$C_{storage} = C_{ref} \left(\frac{M_b}{S_{ref}}\right)^n, \text{ where } M_b = M_{CO_2} \frac{D_b}{365}$$

Oversize pipeline:

$$CAPEX_{per\ ton} = CAPEX_{per\ ton\ temp} + CAPEX_{rel,per\ ton}[€/t]$$
$$OPEX_{per\ ton} = OPEX_{per\ ton\ temp} + OPEX_{rel,per\ ton}[€/t]$$



Cost of direct emissions:

Only operational costs accounting for the carbon price

Model

Final Cost

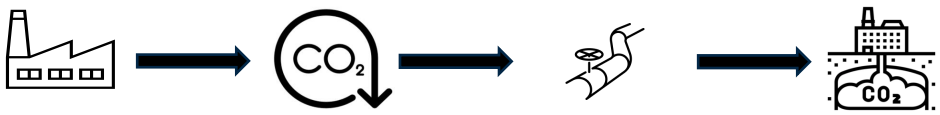
$$M_{CO_2,fail} = M_{CO_2} \frac{D_f}{365}$$

$$CAPEX_{per\ ton,resilience,j} = CAPEX_{per\ ton,j} \frac{M_b}{M_{CO_2,fail}}$$

Operational costs unaffected

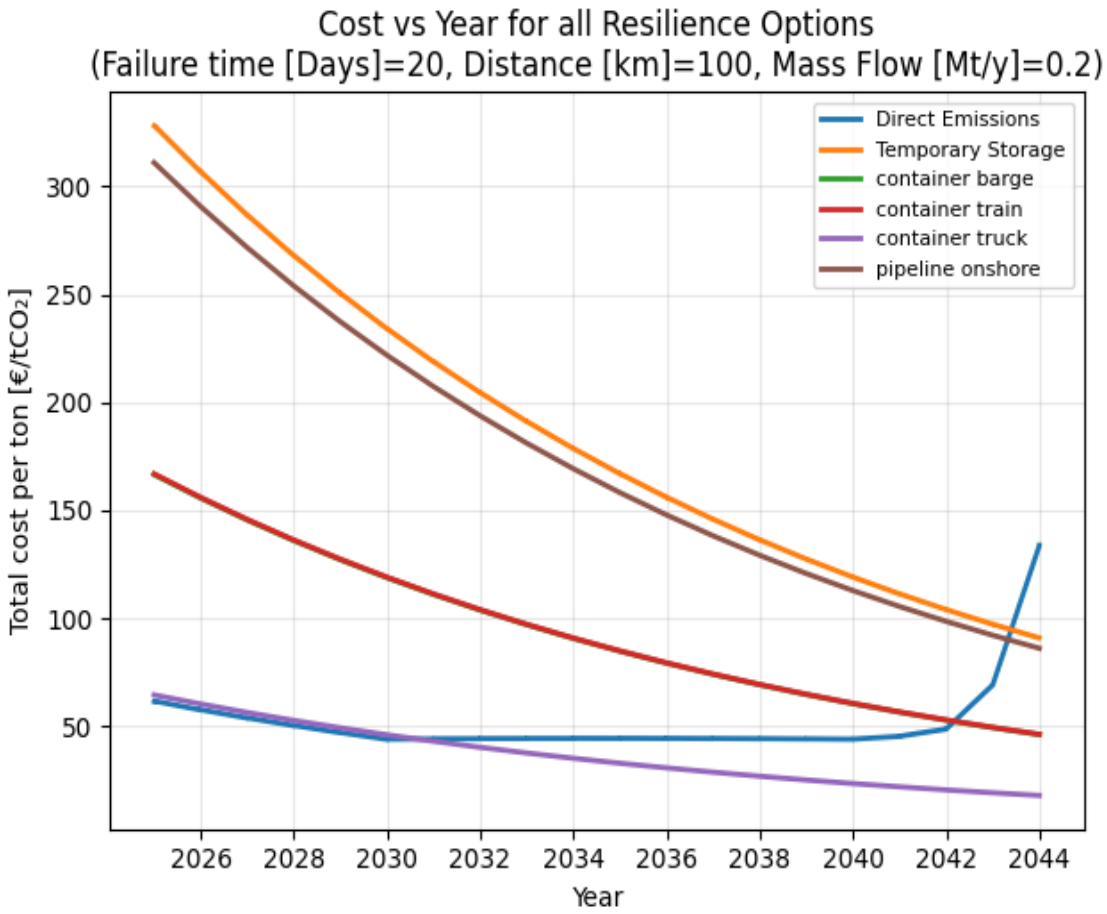
$$C_{t,PV,i} = \frac{CAPEX_{per\ ton,resilience,j} + OPEX_{per\ ton,resilience,j}}{(1+r)^{(t-t_0)}}$$

Influence of the length of Connection

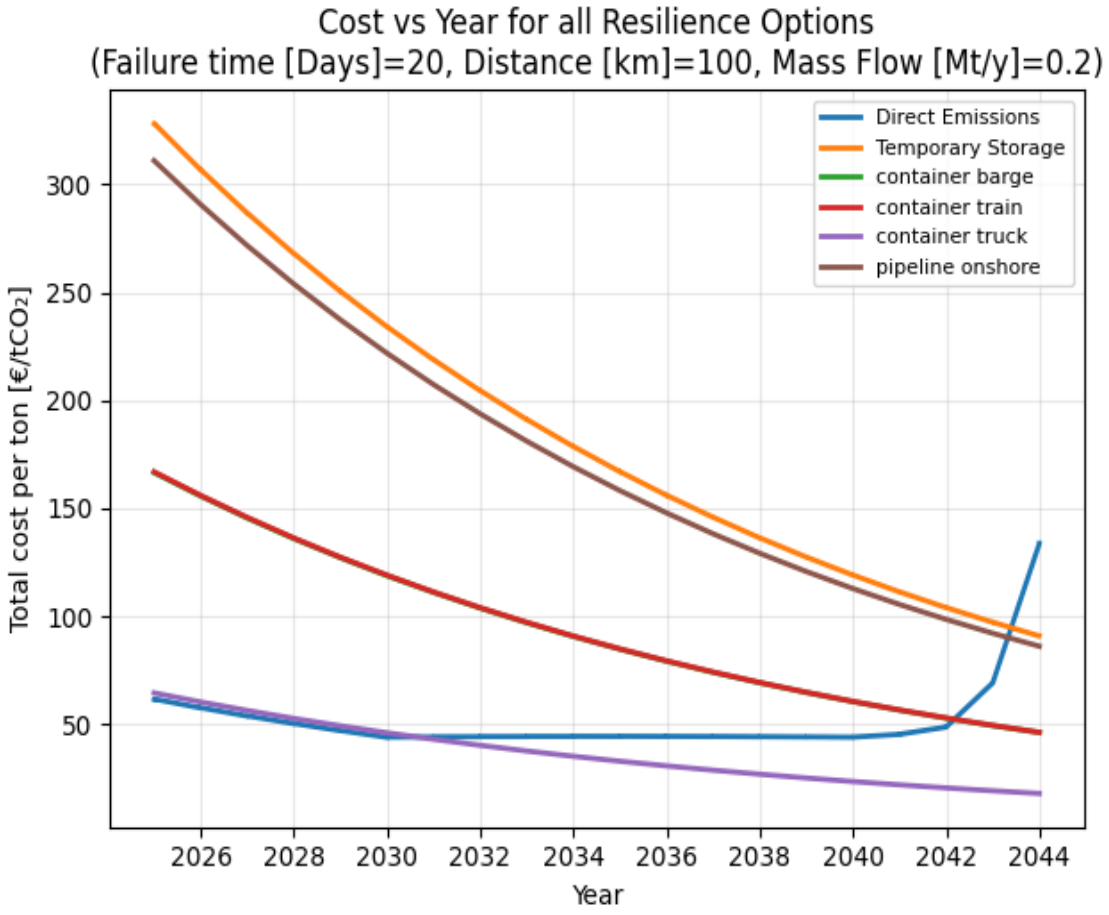


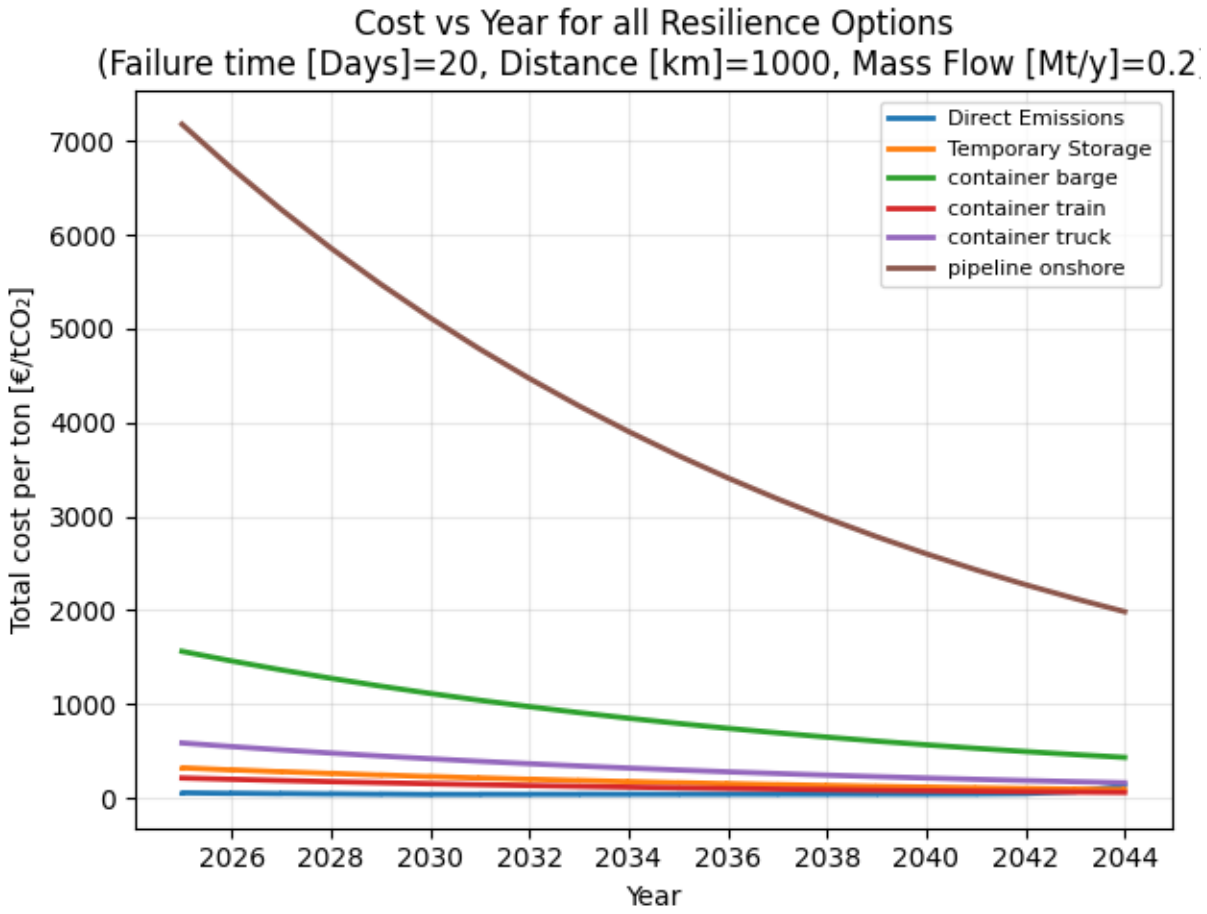
Toy model:

- One emitter connected to the storage site
- Variation of input parameters
- Total cost for all resilience option aggregated throughout the years

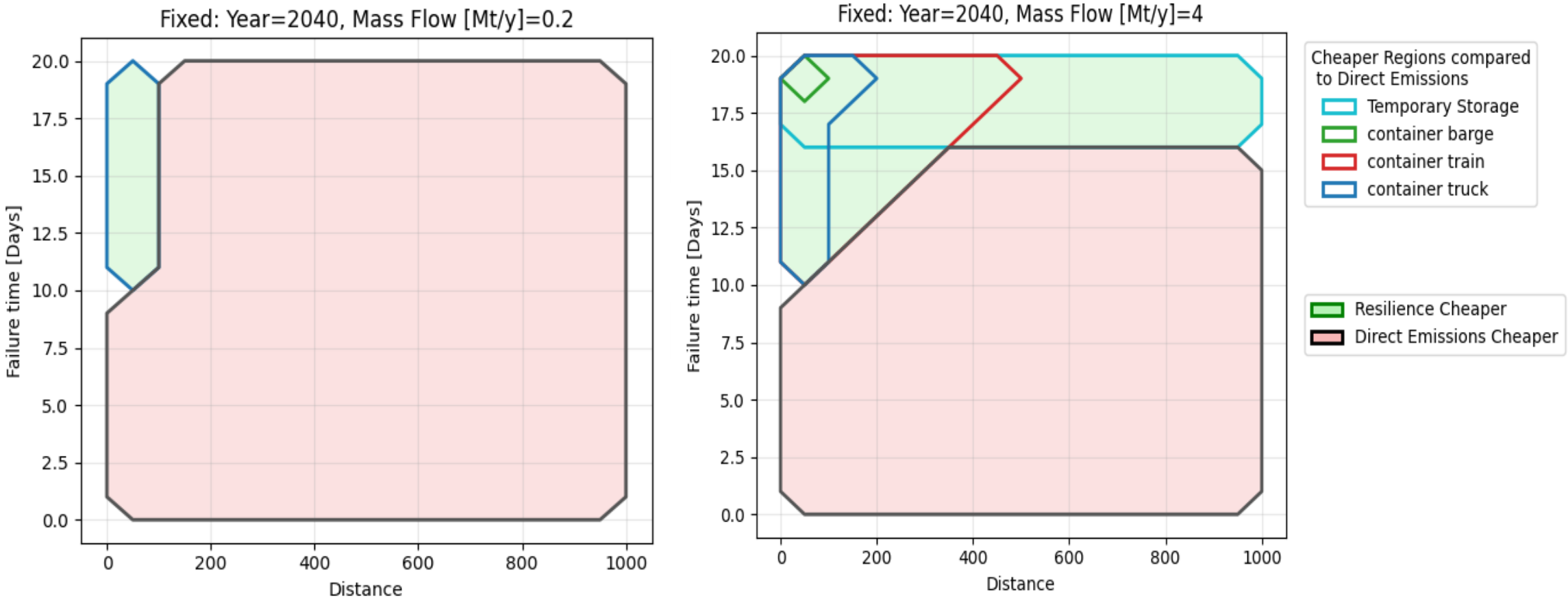


Influence of the length of Connection

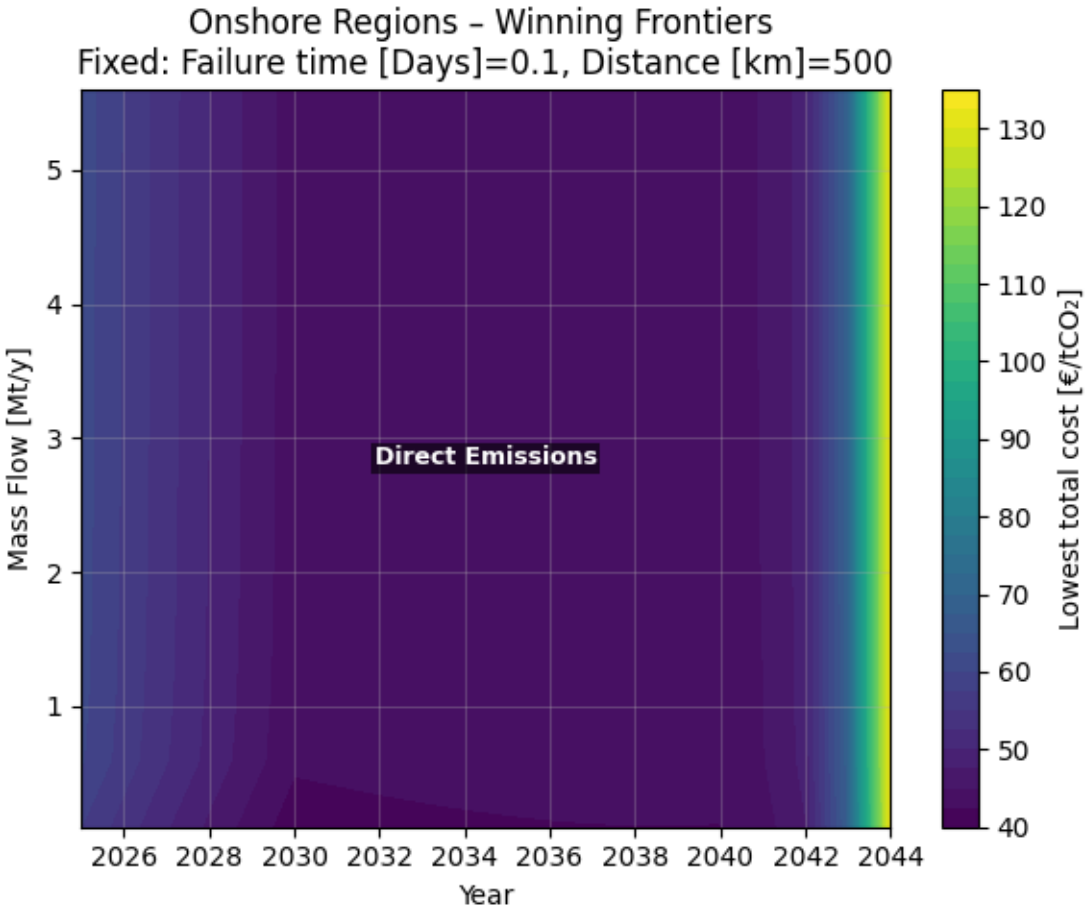


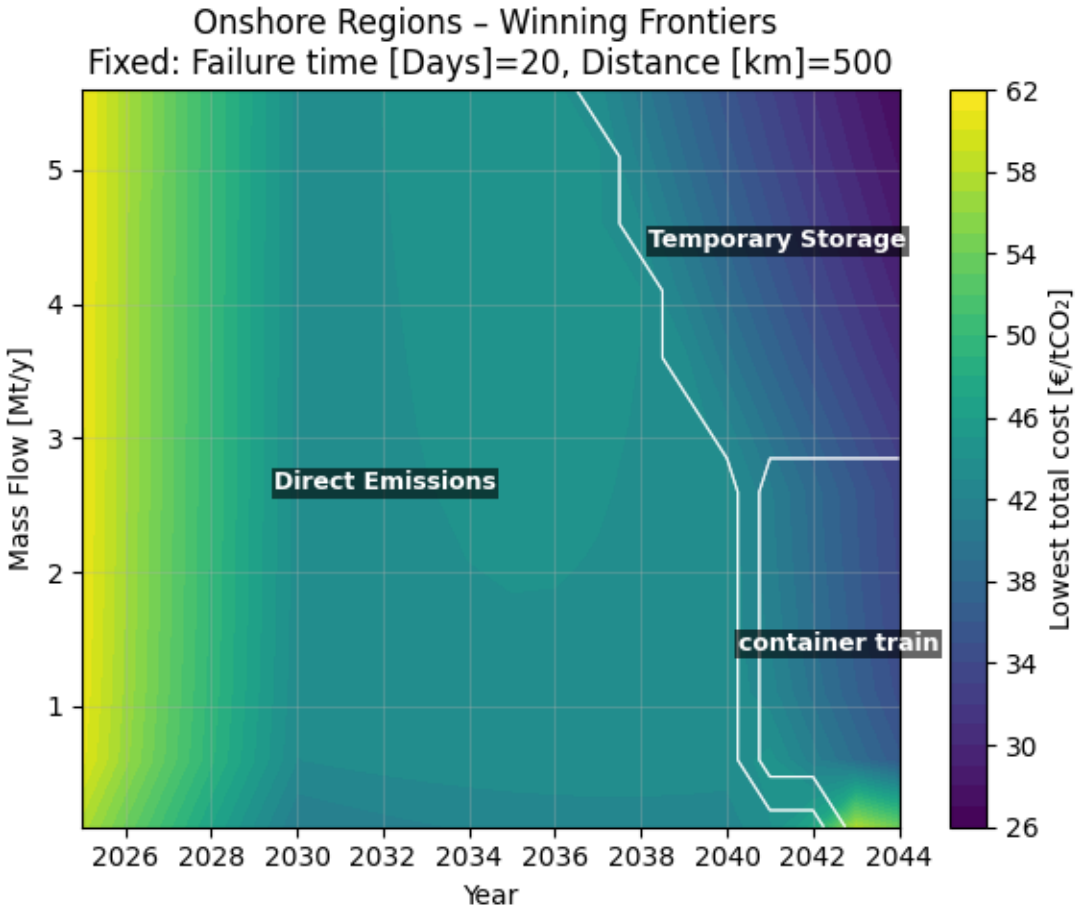


WtE power plant Mass Flow

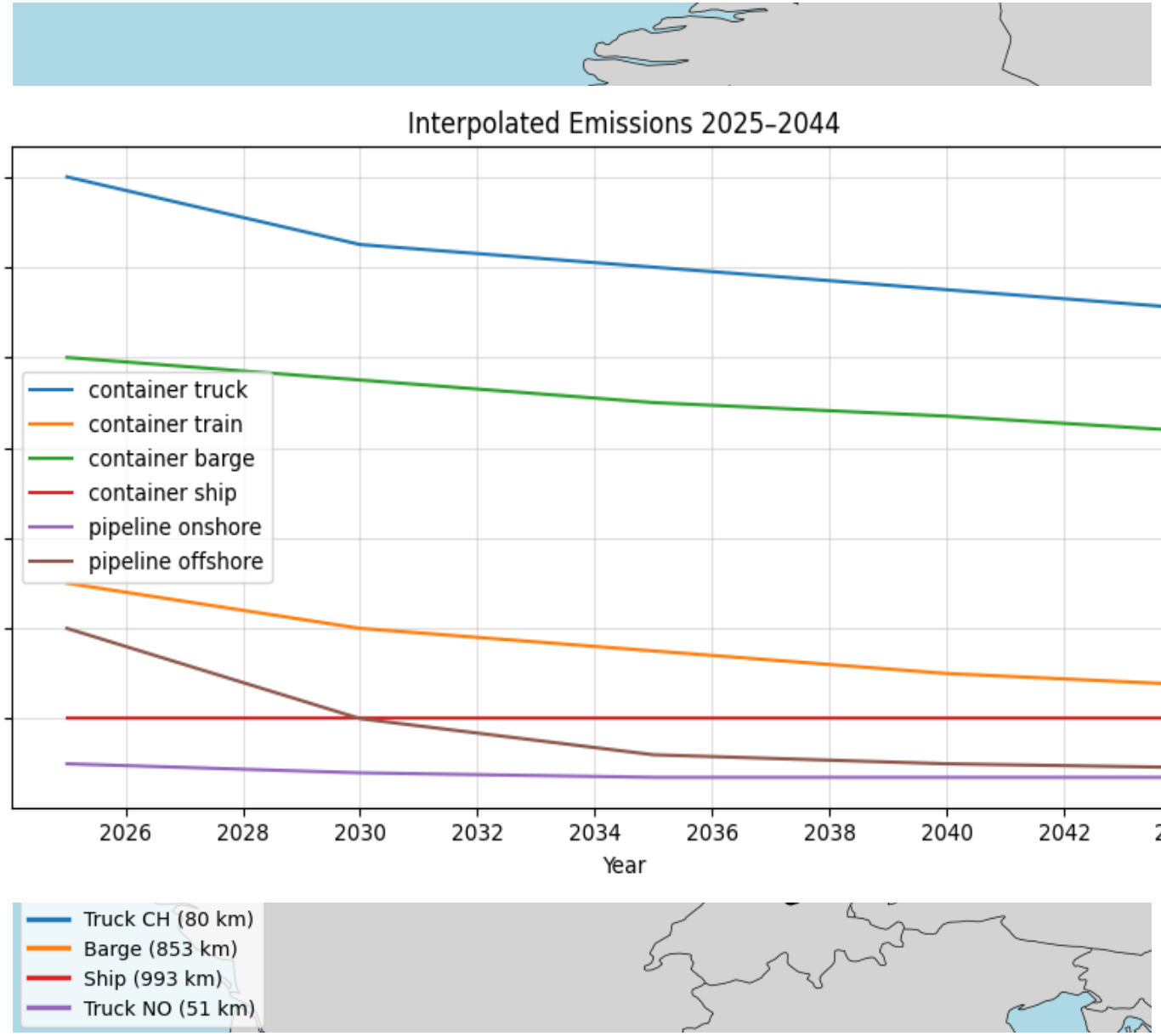


Main Connection Failure Duration in a Year

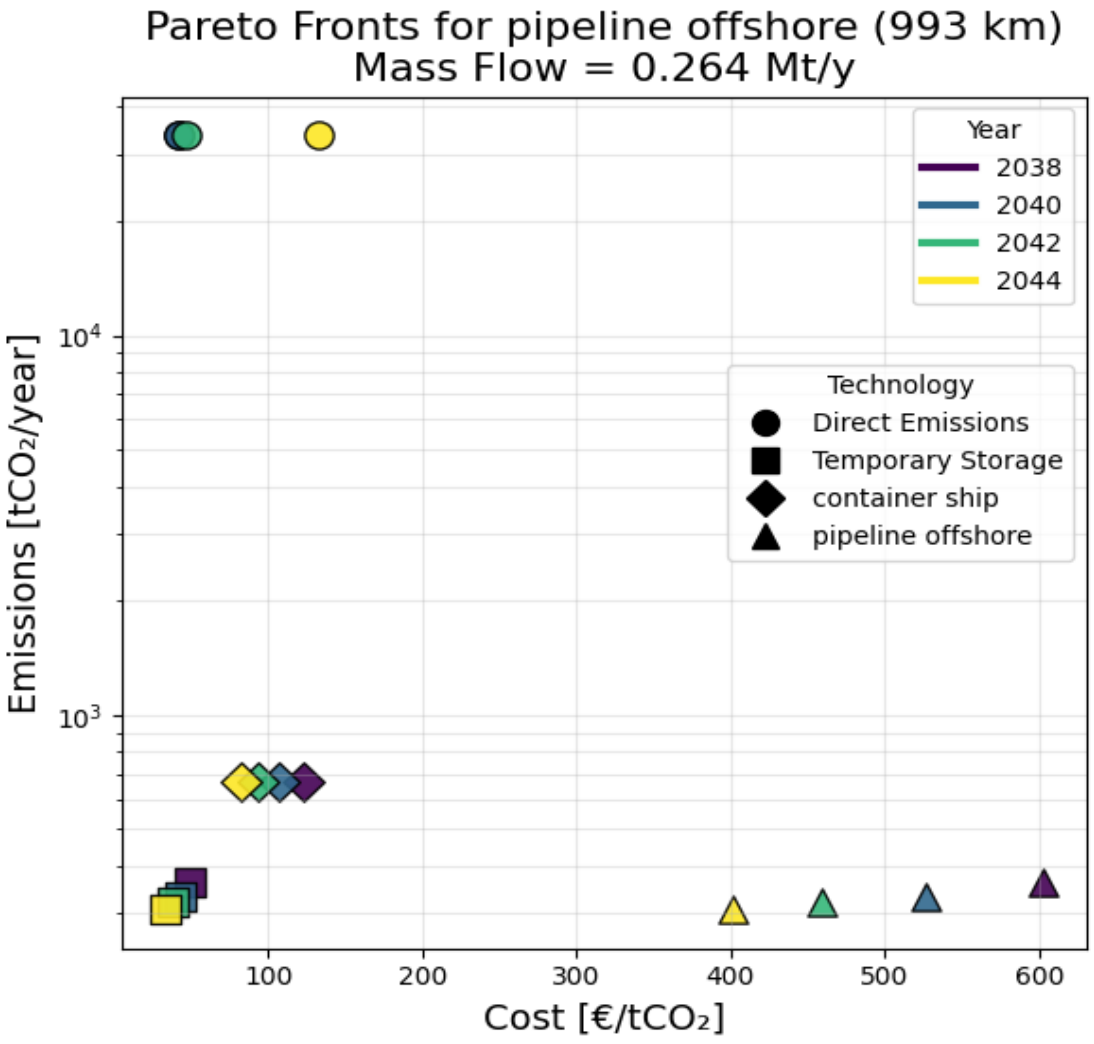




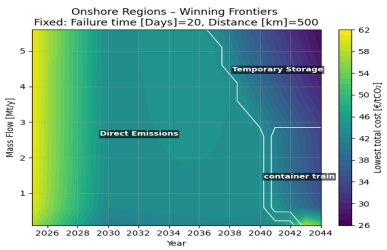
Hagenholz Case Study



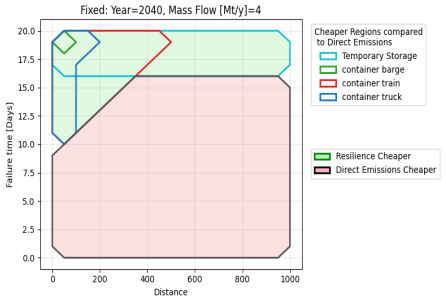
Hagenholz Case Study Results



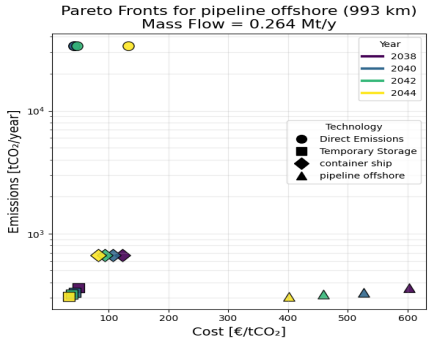
Conclusions



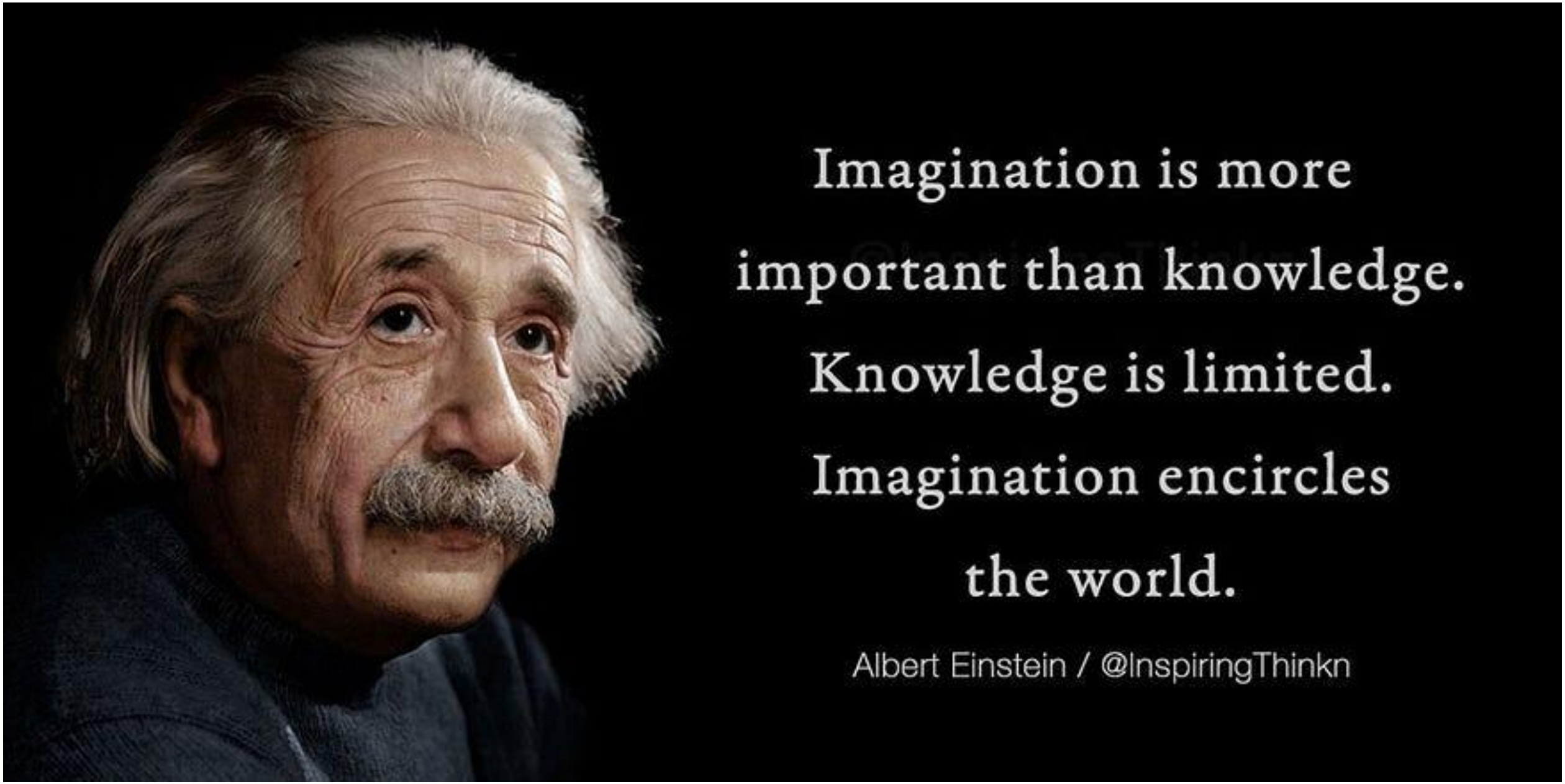
Biggest influence on the cost of options from high mass flow and higher failure rates, due to higher utilization



- Temporary storage seems to be the most favorable option, especially at higher distances and CO_2 flows
- Trucks and trains as good alternatives for onshore regions and ships for offshore regions



Direct Emissions are mostly the cheapest, however a trade off between emissions and cost emerges


Imagination is more
important than knowledge.
Knowledge is limited.
Imagination encircles
the world.
Albert Einstein / @InspiringThinkn

Appendices: Truck and train model

$$t_{\text{trip}} = 2t_{\text{load}} + t_{\text{border}} + t_{\text{break}} + \frac{2L}{v} \text{(Oeuvray et al. 2024)[h]}$$

$$t_{\text{avail}} = D_b \times 8 \quad \text{[h]}$$

$$N_{\text{trips}} = \max\left(1, \left\lfloor \frac{t_{\text{avail}}}{t_{\text{trip}}} \right\rfloor\right) \text{(Oeuvray et al. 2024)}$$

$$M_b = M_{\text{CO}_2} \frac{D_b}{365} \text{[t]}$$

$$N_{\text{truck}} = \min\left\{ n \in \mathbb{Z} \,\middle|\, n \geq 1 \text{ and } n \geq \frac{M_b}{N_{\text{trips}} Cap_{\text{truck}}} \right\}$$

Appendices: Modeling of Resilient Options

For barges and ships:

$$UC = a_1 + \frac{a_2}{L} + \frac{a_3}{M_{CO_2}} [€/t{\cdot}km]$$

For pipelines:

$$UC = a_1 + a_2 \left(\frac{L}{d_{ref}}\right)^{a_3} \left(\frac{M_{CO_2}}{m_{ref}}\right)^{a_4} [€/t{\cdot}km]$$